Effat University — College of Engineering
Department of Computer Science

CS3081 Artificial Intelligence

LAB 3: Propositional Logic & Model Checking

Group Challenge

Instructor	Dr. Naila Marir	**Duration**	1 hour 30 minutes
Semester	Fall 2026	**Group size**	3–4 students
Submission	One GitHub link per group	**Total marks**	100

Group No.			
Member name	**Student ID**	**Member name**	**Student ID**
Nafisa Mahmudulhasan	S24109378		
Basma Elfaki	S24109310		

1. Objectives

1. Represent facts as propositional logic sentences using Not, And, Or and Implication.
2. Explain how `model_check` decides whether a knowledge base entails a query.
3. Encode a new logic puzzle in Python and solve it with model checking.

2. Tools and Files

1. Python 3 and the package `termcolor` (install with `pip install termcolor`).
2. Lab files (on Blackboard): `logic.py`, `harry.py`, `clue.py`, `puzzle.py`, `mastermind.py`. Put all of them in one folder.
3. A GitHub account for each group.

3. Lab Plan

Step	Activity	Time
0	Setup: run harry.py and clue.py	5 min
1	Task 1 — Understand the code	20 min
2	Task 2 — Change the knowledge	20 min
3	Task 3 — Knights & Knaves challenge	30 min
4	Write the PDF and push to GitHub	15 min

4. Task 1 — Understand the Code (30 marks, 20 min)

1. **harry.py:** Write the truth table for `rain`, `hagrid`, `dumbledore` (8 rows). Mark the rows where the knowledge base is true. Use the table to explain why the program prints `True`.

Rain	Hagrid	Dumbledore	KB
F	F	F	F
F	F	T	F
F	T	F	F

F	T	T	F
T	F	F	F
T	F	T	T
T	T	F	F
T	T	T	F

Why?

The knowledge base is true in only one row, and in that row rain is true. Since rain is true whenever the knowledge base is true, the knowledge base entails rain, so the program prints True.

2. **clue.py:** The program prints MsScarlet, library and knife. Explain in plain English why each one must be true.

MsScarlet: It's not Mustard and not Plum, so it's Scarlet.
Library: It's not the kitchen and not the ballroom, so it's the library.
Knife: It's not the revolver. The guess Scarlet, library, wrench was wrong, and since Scarlet and the library are correct, the wrench is wrong. So it's the knife.

5. Task 2 — Change the Knowledge (30 marks, 20 min)

For each change: **(a)** predict the result, **(b)** run the code, **(c)** explain the result.

1. **harry.py:** Delete the last fact `dumbledore` from the knowledge base. What does the program print now? Why?

   a) Prediction: False

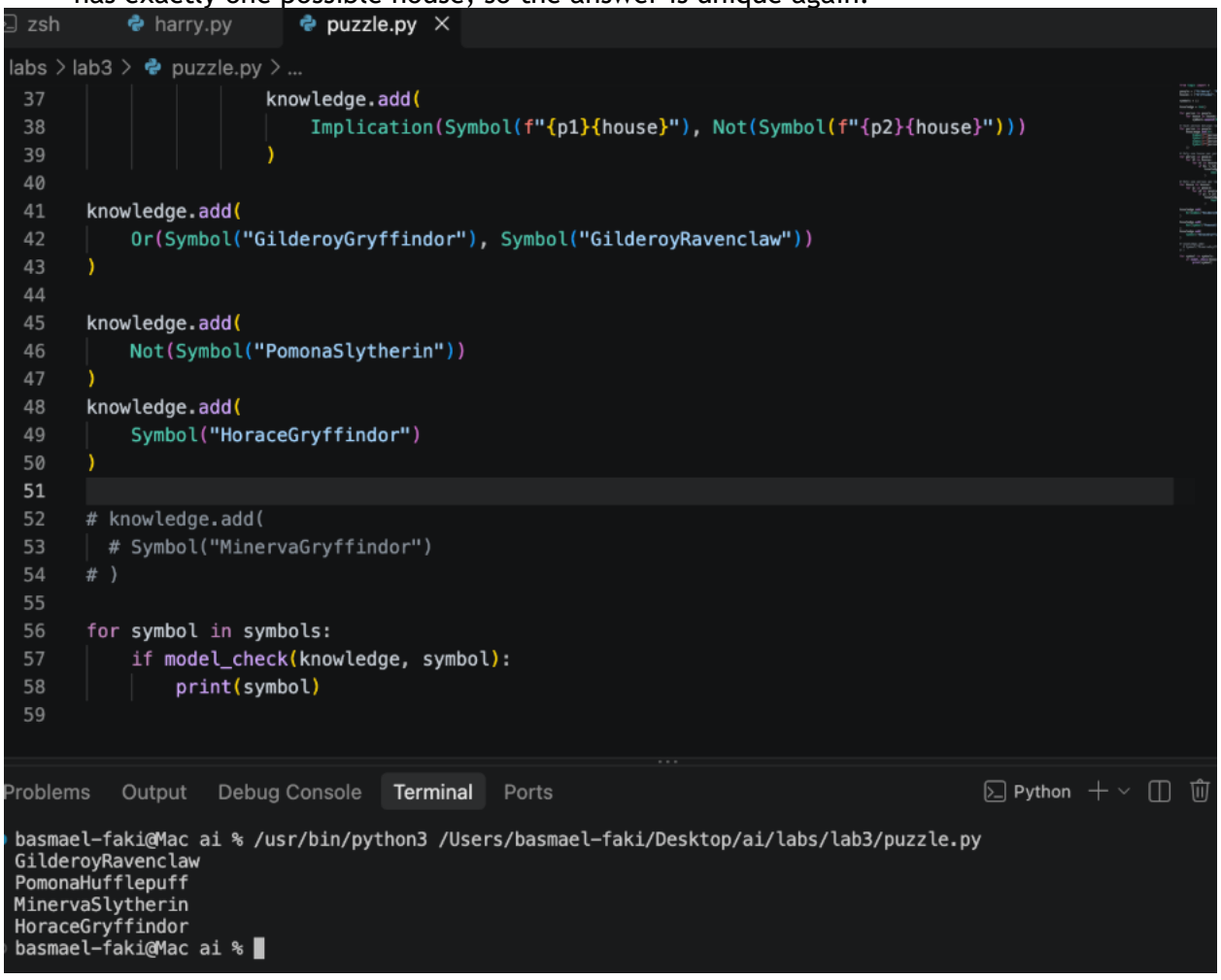
   b) Result: False

   c) Explanation: Without the fact dumbledore, there are now cases where the knowledge base is true but rain is false. So the program can't be sure it rained, and it prints False.

2. **puzzle.py:** Delete the clue `MinervaGryffindor`. Who can still be placed in a house? Then add **one new clue of your own** that makes the answer unique again.

a) Prediction: No one will be placed.
b) Result: Nothing is printed, so no one can be placed.
c) Explanation: Without this clue, the remaining clues allow many possible arrangements. No person is in the same house in all of them, so the program can't be sure of anyone's house.

- **New clue:** Horace is in Gryffindor.
- **Explanation:** Horace takes Gryffindor, so Gilderoy must be in Ravenclaw. Pomona can't be in Slytherin, so she goes to Hufflepuff. Minerva gets the last house, Slytherin. Now everyone has exactly one possible house, so the answer is unique again.



6. Task 3 — Group Challenge: Knights & Knaves (30 marks, 30 min)

On an island, a **knight** always tells the truth and a **knave** always lies. Every person is one or the other, never both. Create a new file `knights.py` that imports `logic.py` and solves the puzzles below.

Hint — use this pattern for what a person says:

```
Implication(AKnight, statement)      # a knight's statement is true
Implication(AKnave, Not(statement))  # a knave's statement is false
```

1. **Puzzle 0:** A says "I am both a knight and a knave."
2. **Puzzle 1:** A says "We are both knaves." B says nothing.
3. **Puzzle 2:** A says "We are the same kind." B says "We are of different kinds."

zsh harry.py puzzle.py knights.py

labs > lab3 > knights.py > ...

```python
from logic import *

AKnight, AKnave = Symbol("A is a Knight"), Symbol("A is a Knave")
BKnight, BKnave = Symbol("B is a Knight"), Symbol("B is a Knave")
CKnight, CKnave = Symbol("C is a Knight"), Symbol("C is a Knave")

A = And(Or(AKnight, AKnave), Not(And(AKnight, AKnave)))
B = And(Or(BKnight, BKnave), Not(And(BKnight, BKnave)))
C = And(Or(CKnight, CKnave), Not(And(CKnight, CKnave)))

knowledge0 = And(A,
    Implication(AKnight, And(AKnight, AKnave)),
    Implication(AKnave, Not(And(AKnight, AKnave))))

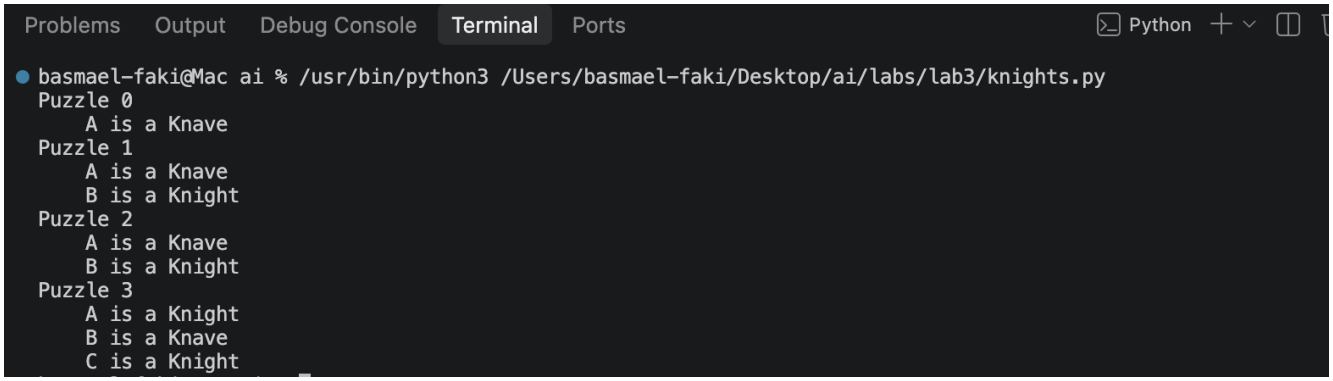
knowledge1 = And(A, B,
    Implication(AKnight,
    Implication(AKnave, Not(And(AKnave, BKnave)))))

same = Or(And(AKnight, BKnight), And(AKnave, BKnave))
knowledge2 = And(A, B,
    Implication(AKnight, same),
    Implication(AKnave, Not(same)),
    Implication(BKnight, Not(same)),
    Implication(BKnave, same))

saidKnight = And(Implication(AKnight, AKnight), Implication(AKnave, Not(AKnight)))
saidKnave = And(Implication(AKnight, AKnave), Implication(AKnave, Not(AKnave)))
knowledge3 = And(A, B, C,
    Or(saidKnight, saidKnave),
    Implication(BKnight, saidKnave),
    Implication(BKnave, Not(saidKnave)),
    Implication(BKnight, CKnave),
    Implication(BKnave, Not(CKnave)),
    Implication(CKnight, AKnight),
    Implication(CKnave, Not(AKnight)))

symbols = [AKnight, AKnave, BKnight, BKnave, CKnight, CKnave]
puzzles = [knowledge0, knowledge1, knowledge2, knowledge3]

for i in range(4):
    print("Puzzle", i)
    for symbol in symbols:
        if model_check(puzzles[i], symbol):
            print("   ", symbol)
```



Bonus (+10 marks) — Puzzle 3: A says either "I am a knight." or "I am a knave." (you don't know which). B says "A said 'I am a knave'." B also says "C is a knave." C says "A is a knight."

7. Submission

Create **one public GitHub repository** named `Lab3_GroupX` and submit **only its link** on Blackboard. The repository must contain:

1. `Lab3_GroupX.pdf` — group number, member names, and for each task: your answer, your code, and a screenshot of the output.
2. All `.py` files — `logic.py`, your changed `harry.py` and `puzzle.py`, and your new `knights.py`.
3. `README.md` — member names and one line on how to run `knights.py`.

Before you submit: open your link in a private (incognito) browser window to check that it works.

8. Grading Rubric

Task	What is checked	Marks
Task 1	Correct truth table and clear explanations	30
Task 2	Predictions, results and explanations; a working new clue	30
Task 3	Puzzles 0–2 encoded correctly with output and explanation	30
Submission	Clear PDF and complete GitHub repository	10
Total		**100**
Bonus	Puzzle 3 solved and explained	+10